

Functional Equations for Beginners

An olympiad introduction

Zhandaulet · 30 September 2026

What this handout covers: elementary substitutions; injectivity and surjectivity; destroying f 's and symmetry (including bumps); functional equations on $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$; Cauchy's equation over $\mathbb{Q}$ and over $\mathbb{R}$; Jensen; heuristics; practice problems. *How to use it:* try every example yourself before reading the solution. Exercises marked as such are left for you on purpose.

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1. Introduction

A **function** $f : A \rightarrow B$ is a rule that assigns to every element $x \in A$ exactly one element $f(x) \in B$.

- The set A is the **domain** of f .
- The set B is the **codomain**.
- The set $\{f(x) : x \in A\}$ is the **image** (or range) of f .

A **functional equation** is an equation that relates the values of one or more unknown functions at different points. Solving it means finding *all* functions that turn the given relation into an identity (for all allowed values of the variables).

Example 1.1. Typical shapes:

$$\begin{aligned} f(x+y) + f(x-y) &= 2f(x) + f(y), \\ f(xy) &= f(x)f(y), \\ f(x+y) &= f(x) + f(y). \end{aligned}$$

Always start by guessing. Before any formal work, ask yourself:

- Do constant functions work?
- Do linear functions $f(x) = cx + d$ work?
- Do quadratic polynomials work?
- Does the zero function work?

Knowing a plausible family of solutions tells you what properties you should try to prove. For instance, if $f(x) = x$ seems to work, you will probably *not* be able to prove that f is constant.

Exercise 1.2. For each of the following equations invent *one* function that satisfies it. Use a **different** function for each item.

- $f(x) = f(x+2)$
- $f(xy) = f(x)f(y)$
- $f(x+y) = f(x) + f(y)$
- $f(x+y) + f(x-y) = 2f(x) + f(y)$

2. Elementary Substitutions and Observations

Most solutions begin with simple substitutions: set some variables to 0, set two variables equal, set one variable to the negative of another, etc.

First moves. Whenever the domain permits, try

$$x = 0, \quad y = 0, \quad x = y, \quad y = -x, \quad y = 1, \quad y = f(x)$$

(and any other simple values that make an expression inside f collapse).

Example Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(x + y)(f(x) - f(y)) = (x - y)(f(x) + f(y)) \quad \forall x, y \in \mathbb{R}.$$

Expand both sides:

$$xf(x) - xf(y) + yf(x) - yf(y) = xf(x) + xf(y) - yf(x) - yf(y).$$

Simplifying yields

$$xf(y) = yf(x) \quad \forall x, y \in \mathbb{R}. \quad (1)$$

Method I (ratio). If $x \neq 0$ and $y \neq 0$, then

$$\frac{f(x)}{x} = \frac{f(y)}{y}.$$

Both sides equal some constant c , so $f(x) = cx$ for $x \neq 0$. Setting $x = 0$ in (1) gives $f(0) = 0$. Thus $f(x) = cx$ for all x . **Method II (clever substitution).** Simply put $y = 1$ in (1):

$$xf(1) = f(x) \quad \forall x \in \mathbb{R}.$$

Writing $c = f(1)$ we obtain $f(x) = cx$ at once. Verification shows that every function of the form $f(x) = cx$ works.

Never forget the final verification. After you obtain a candidate family of functions, plug them back into the original equation and check which (if any) constants or parameters survive.

Example Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy) = \frac{f(x) + f(y)}{x + y} \quad \text{for all } x, y \text{ with } x + y \neq 0.$$

Call the given equation $P(x, y)$. Set $y = 0$ (allowed when $x \neq 0$):

$$f(0) = \frac{f(x) + f(0)}{x} \implies f(x) = xf(0) - f(0) = c(x - 1), \quad x \neq 0,$$

where $c = f(0)$. Now set $y = x \neq 0$:

$$f(x^2) = \frac{2f(x)}{2x} = \frac{f(x)}{x}.$$

If $f(x) = c(x - 1)$, comparing both sides forces $c = 0$. Hence $f \equiv 0$ on $\mathbb{R} \setminus \{0\}$, and also $f(0) = 0$. The zero function satisfies the original equation.

How to guess the form. If an expression looks homogeneous of degree 1, try linear functions first. If it looks multiplicative, try $f \equiv 0$, power functions x^k , etc. Knowing a plausible family makes the subsequent formal work much easier.

3. Injectivity and Surjectivity

Many functional equations cannot be solved by substitutions alone. Two properties of a function often become the key that unlocks the rest of the solution.

Definition 3.1. A function $f : A \rightarrow B$ is **injective** (one-to-one) if $f(a) = f(b)$ implies $a = b$. A function $f : A \rightarrow B$ is **surjective** (onto) if for every $y \in B$ there exists at least one $x \in A$ such that $f(x) = y$.

How to prove surjectivity — isolated pieces. Scan the equation and ask, for each variable:

- Does it appear on only one side, and never inside an f ?
- Does it appear inside only one particular $f(\dots)$, while the rest of the equation treats it simply (or not at all)?

Any variable that answers “yes” is *isolated*. Fix every other variable and let the isolated one run through the whole domain. If the side that does not contain the outer f can take every real value, then the image of f must be all of $\mathbb{R}$. When every occurrence of a variable y is already wrapped inside an f , surjectivity is especially pleasant: the substitution “set $f(y) = t$ ” becomes legitimate for every real t .

Example Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - f(y)) = 1 - x - y \quad \forall x, y \in \mathbb{R}.$$

The variable y on the right-hand side is completely free of f . Fix any x and let y vary over $\mathbb{R}$: the right-hand side runs over all of $\mathbb{R}$. The left-hand side is always a value of f , therefore f is surjective. Consequently there exists some y_0 with $f(y_0) = 0$. Substitute $y = y_0$:

$$f(x) = 1 - x - y_0 = -x + c, \quad c = 1 - y_0.$$

Plug $f(t) = -t + c$ back into the original equation:

$$\begin{aligned} \text{LHS} &= f(x - (-y + c)) = f(x + y - c) = -(x + y - c) + c = -x - y + 2c, \\ \text{RHS} &= 1 - x - y. \end{aligned}$$

Equality forces $2c = 1$, i.e. $c = \frac{1}{2}$. Thus $f(x) = -x + \frac{1}{2}$. Direct verification confirms that this function works.

How to prove injectivity. Again look for an isolated variable. Suppose $f(y_1) = f(y_2)$. Substitute the pairs (x, y_1) and (x, y_2) into the equation. The parts that contain $f(y)$ stay the same, while the isolated variable produces y_1 on one side and y_2 on the other. Equality forces $y_1 = y_2$.

Example Suppose

$$f(f(y) + x f(x)) = y + f(x)^2 \quad \forall x, y \in \mathbb{R}.$$

Show that f is bijective.

Keep x fixed and let y vary. The right-hand side contains the free term y , so it runs through all of $\mathbb{R}$. The left-hand side is always a value of f , therefore f is surjective. Surjectivity supplies a point x_0 with $f(x_0) = 0$. Substituting that particular x yields

$$f(f(y)) = y.$$

Hence f is bijective and is its own inverse — a far stronger statement than surjectivity alone.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x + f(y)) = f(x) + y \quad \forall x, y \in \mathbb{R}.$$

Prove that f is injective, and outline how to find all such functions.

Injectivity. Suppose $f(a) = f(b)$. Fix an arbitrary x and write the equation for $y = a$ and for $y = b$:

$$f(x + f(a)) = f(x) + a,$$

$$f(x + f(b)) = f(x) + b.$$

The left-hand sides are identical, therefore $f(x) + a = f(x) + b$, so $a = b$. Hence f is injective.

Finding the form. Set $x = 0$:

$$f(f(y)) = f(0) + y.$$

The right-hand side can be any real number, so f is also surjective. In particular there exists c with $f(c) = 0$. Setting $y = c$ in the original equation forces $c = 0$, and we obtain $f(f(y)) = y$. In other words f is an involution: it is its own inverse. From $f(f(y)) = y$ and the original equation one can deduce that f is additive, or simply guess the linear form $f(x) = x + k$ or $f(x) = -x + k$ and check. The important point is that injectivity together with $f(f(y)) = y$ already gives enormous control.

Applying the function to itself again. Whenever you know a relation of the form $f(f(x)) = g(x)$ for some simpler function g , apply f once more. There are two ways to associate the triple composition:

$$f(f(f(x))) = f(g(x)),$$

$$f(f(f(x))) = g(f(x)).$$

Equating them produces a new identity that often involves only a single outer f . For example, if $f(f(x)) = x + 2$, then

$$f(x + 2) = f(f(f(x))) = f(x) + 2.$$

The special case $f(f(x)) = x$ (an *involution*) is even cleaner: f is its own inverse, every double application may be removed, and you may solve for the inner variable whenever an outer f appears.

Once you know that f is injective you may “cancel f ” on both sides of an equality:

$$f(A) = f(B) \implies A = B.$$

This is one of the most frequently used consequences of injectivity.

4. Two Important Ideas

4.1. Destroying the f 's

When both sides of the equation contain expressions of the form $f(\dots)$, try to make two of those inner expressions equal by a suitable substitution. Then the outer f 's cancel (or give a simpler relation).

Example Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) = f(x^2 + 2y) + f(x) \quad \forall x, y \in \mathbb{R}.$$

We want the arguments of two f 's to coincide. Set the left argument equal to the first right argument:

$$x + y = x^2 + 2y \iff y = x - x^2.$$

Substitute $y = x - x^2$:

$$\begin{aligned} f(x + (x - x^2)) &= f(x^2 + 2(x - x^2)) + f(x) \\ f(2x - x^2) &= f(2x - x^2) + f(x) \\ \implies f(x) &= 0 \quad \text{for all } x. \end{aligned}$$

The zero function satisfies the original equation.

The same idea works with any pair of arguments. Sometimes the substitution that equates two arguments produces a polynomial equation of odd degree; such a polynomial always has a real root, which lets you reach every real number and conclude that f vanishes everywhere.

4.2. Symmetry and bumps

If one side of the equation is symmetric in x and y while the other is not, write the equation once as $P(x, y)$ and once as $P(y, x)$. Comparing the two forms often produces a useful relation.

Example Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x + f(x) + 2f(y)) = x + f(x) + y + f(y) \quad \forall x, y \in \mathbb{R}.$$

- Prove that f is injective.
- Find all such functions.

(a) Injectivity. Set $x = 0$:

$$f(2f(y)) = f(0) + y + f(y). \quad (*)$$

Suppose $f(y_1) = f(y_2)$. Then the left-hand side of $(*)$ is the same for y_1 and y_2 , therefore

$$f(0) + y_1 + f(y_1) = f(0) + y_2 + f(y_2) \implies y_1 = y_2.$$

Hence f is injective. **(b) Finding the form.** The right-hand side of the original equation is completely symmetric in x and y . Write the equation for the pair (y, x) :

$$f(y + f(y) + 2f(x)) = y + f(y) + x + f(x).$$

The two right-hand sides are identical, so the two left-hand sides are equal. By injectivity the arguments are equal:

$$x + f(x) + 2f(y) = y + f(y) + 2f(x) \implies f(x) - x = f(y) - y.$$

Both sides equal a constant c , so $f(x) = x + c$. Substituting back into the original equation forces $c = 0$. Thus $f(x) = x$.

Almost-symmetric equations and their bumps. Many equations are symmetric in x and y *except* for one or two terms. Those exceptional terms are the **bumps**. Far from being a nuisance, they are usually the most useful part of the equation. The systematic reaction is:

1. Write the original statement as $P(x, y)$.
2. Write the swapped statement $P(y, x)$.
3. Cancel every term that appears in both.

Whatever remains involves only the asymmetric pieces. That residual relation is frequently exactly what is needed to prove that $f(x) - x$ is constant, or that two expressions inside f must be equal. Resist the temptation to “clean up” the equation by removing asymmetric terms at the outset — those terms are often the only reason the comparison $P(x, y)$ versus $P(y, x)$ produces anything non-trivial.

5. Functional Equations on $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$

When the domain is the positive integers, the integers, or the rationals, the situation changes in two important ways:

1. There is no “pathological” solution coming from a Hamel basis; every additive function $\mathbb{N} \rightarrow \mathbb{N}$ (or $\mathbb{Z} \rightarrow \mathbb{Z}$) that satisfies mild extra conditions is linear.
2. One can often use induction, number-theoretic arguments (divisibility, prime factorization), or the well-ordering of $\mathbb{N}$.

Example Find all $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(m + n) + f(m - n) = 2f(m) + 2f(n) \quad \text{for all } m > n \geq 1.$$

Set $n = 1$ and write $a_m = f(m)$. The recurrence becomes

$$a_{m+1} + a_{m-1} = 2a_m + 2a_1.$$

This is a linear non-homogeneous recurrence. The homogeneous solution is linear; a particular solution is quadratic. After fitting the first two values one obtains $f(n) = cn^2$. Direct substitution shows that every such function works (provided c is a positive integer so that the image stays inside $\mathbb{N}$).

On $\mathbb{N}$ and $\mathbb{Z}$ the following moves are especially productive:

- Replace x by $x + 1$ (or set one variable equal to 1). This almost always produces a recurrence that relates the value at a point to the values at nearby points. Once a few initial values are known, the recurrence determines the function completely.
- Use induction on the size of the argument. The logical trigger is the same as in any other olympiad problem: you must be able to express a new value in terms of values already controlled.
- If the equation looks multiplicative, evaluate it at primes and then extend by unique factorisation.
- On $\mathbb{N}$, well-ordering can sometimes replace induction: consider a minimal counter-example and derive a contradiction.

In short, the discrete setting gives you induction and recurrences “for free”; the continuous setting does not.

6. Cauchy’s Functional Equation

A huge number of olympiad functional equations reduce to Cauchy’s equation (or a close relative). We treat the equation carefully first over $\mathbb{Q}$, then over $\mathbb{R}$.

6.1. Cauchy’s equation over $\mathbb{Q}$

For this subsection all functions are $f : \mathbb{Q} \rightarrow \mathbb{Q}$. We highly encourage the reader to try these examples on their own before reading the solutions; they are good practice problems!

Example (Cauchy over $\mathbb{Q}$) Solve $f(x + y) = f(x) + f(y)$ over $\mathbb{Q}$.

Trying out the most general linear function $f(x) = kx + c$, we find that $c = 0$ but k can be anything. So our guess is that the answer is $f(x) = kx$. The family of solutions has one free parameter. It therefore makes sense to set $k = f(1)$ and express everything else in terms of k .

- Setting $x = y = 0$ gives $f(0) = 0$.
- Setting $x = 1, y = 1$ gives $f(2) = f(1) + f(1) = 2k$.
- Setting $(x, y) = (2, 1)$ gives $f(3) = 3k$, and by induction $f(n) = kn$ for every integer $n \geq 1$.
- For negative integers, put $x = -y$: then $f(x) + f(-x) = 0$, so f is odd. Thus $f(n) = kn$ holds for all $n \in \mathbb{Z}$.
- To reach the rest of $\mathbb{Q}$, first evaluate at halves:

$$f\left(\frac{1}{2}\right) + f\left(\frac{1}{2}\right) = f(1) = k \quad \implies \quad f\left(\frac{1}{2}\right) = \frac{k}{2}.$$

- For a general rational p/q (with $q > 0$) we have

$$\underbrace{f(p/q) + \cdots + f(p/q)}_{q \text{ times}} = f(p) = kp,$$

hence $f(p/q) = k \cdot p/q$.

We conclude that $f(x) = kx$ for all $x \in \mathbb{Q}$.

Remark. The choice of domain $\mathbb{Q}$ is essential: the argument relies on induction to control the values on $\mathbb{Z}$, and then on repeated addition to reach all of $\mathbb{Q}$. The same proof fails for a function $f : \mathbb{R} \rightarrow \mathbb{Q}$. By contrast, the choice of codomain is irrelevant — the identical argument works for $f : \mathbb{Q} \rightarrow \mathbb{R}$.

Example (Jensen over $\mathbb{Q}$) Solve over $\mathbb{Q}$:

$$f(x) + f(y) = 2f\left(\frac{x+y}{2}\right).$$

Preliminary checks show that $f(x) = kx + c$ works for any constants k and c . Because the constant term is free, we may shift the function without changing the equation. Rewrite the given relation as

$$(f(x) - f(0)) + (f(y) - f(0)) = 2\left(f\left(\frac{x+y}{2}\right) - f(0)\right).$$

Set $g(x) = f(x) - f(0)$. Then g satisfies the same equation and, moreover, $g(0) = 0$:

$$g(x) + g(y) = 2g\left(\frac{x+y}{2}\right).$$

Substituting $(x, y) = (t, 0)$ yields $g(t) = 2g(t/2)$. Using this identity we may rewrite the equation as

$$g(x) + g(y) = 2g\left(\frac{x+y}{2}\right) = g(x+y).$$

Thus g satisfies Cauchy's equation over $\mathbb{Q}$. From the previous example we already know that $g(x) = kx$ for some constant k . Therefore

$$f(x) = g(x) + f(0) = kx + c,$$

where $c = f(0)$.

Example (JMO 2015/4) Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x) + f(t) = f(y) + f(z)$$

for all rational numbers $x < y < z < t$ that form an arithmetic progression.

As before, linear functions $f(x) = kx + c$ satisfy the condition. Rewrite the equation as

$$f(a) + f(a+3d) = f(a+d) + f(a+2d)$$

for $d > 0$. This has a lot of terms in it; it would be great if many of them cancelled. Write down the shifted version

$$f(a - d) + f(a + 2d) = f(a) + f(a + d),$$

which follows by replacing a with $a - d$. Adding:

$$f(a - d) + f(a + 3d) = 2f(a + d) \quad \forall a \in \mathbb{Q} \text{ and } d > 0.$$

At this point we are essentially done:

Exercise 6.1. Reduce this to Jensen's functional equation and complete the problem. (Address the case $d = 0$ separately; it is trivial.)

The main idea was that by shifting a little, several terms cancelled. That led back to Jensen's equation.

6.2. Cauchy's equation over $\mathbb{R}$

The problem becomes very different if we replace $\mathbb{Q}$ by $\mathbb{R}$, since induction is no longer valid. Over $\mathbb{R}$ we obtain new pathological ("bad") solutions to Cauchy's equation that were not present before; their construction requires the axiom of choice and never appears in olympiad solutions.

Theorem 6.2 (Cauchy + regularity $\implies$ linear). Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$. Then $f(qx) = qf(x)$ for any $q \in \mathbb{Q}$. Moreover, f is linear if any of the following hold:

- f is continuous on any interval;
- f is bounded (above or below) on any nontrivial interval;
- there exist (a, b) and $\varepsilon > 0$ such that $(x - a)^2 + (f(x) - b)^2 > \varepsilon$ for every x (i.e. the graph of f omits some disk, however small).

In olympiad problems the most common ways to pass from additive to linear are:

- proving a boundedness condition (for instance $f \geq 0$ on an interval), or
- being given continuity of f in the statement of the problem.

It is extremely rare that one needs to prove continuity as an intermediate step.

Example (Field automorphisms of $\mathbb{R}$) Solve over $\mathbb{R}$:

$$f(x + y) = f(x) + f(y) \quad \text{and} \quad f(xy) = f(x)f(y).$$

We claim that the only solutions are $f(x) = x$ and $f(x) = 0$ (both of which work). By the theorem above it is enough to show that f is nonnegative on some nontrivial interval. The multiplicative relation gives

$$f(t^2) = f(t)^2 \geq 0$$

for every t , so f is bounded below on $[0, +\infty)$. Hence $f(x) = cx$ for some constant c . Substituting into the multiplicative equation yields $cxy = (cx)(cy)$, which forces $c \in \{0, 1\}$.

6.3. Variants of Cauchy's equation

Many contest problems lead to one of the following related equations:

- $f(x + y) + f(x - y) = 2f(x)$ (quadratic / d'Alembert type after a change of variables);

- $f(xy) = f(x) + f(y)$ (for $x, y > 0$ this becomes Cauchy after $x = e^u, y = e^v$);
- $f(xy) = f(x)f(y)$ (multiplicative functions);
- $f(x + y) = f(x)f(y)$ (exponential functions).

The same density argument together with continuity (or another regularity condition) usually yields the classical solutions $f(x) = cx^2$, $f(x) = c \log x$, $f(x) = x^c$, $f(x) = e^{cx}$, etc.

7. More Worked Examples

Example Determine all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + y) = f(x^2 - y) + 2yf(x) + x^2 \quad \forall x, y \in \mathbb{R}.$$

Step 1 — Guess. Linear functions $f(x) = x + c$ or $f(x) = -x + c$ are natural candidates. Trying $f(x) = x$ works; trying $f(x) = -x$ does not. So we expect something close to the identity. **Step 2 — Surjectivity.** Fix x and let y vary. The left side is $f(\text{something})$, while the right side contains the term $-y$ (inside $x^2 - y$) plus a linear term in y . The right side can take every real value, therefore f is surjective. **Step 3 — Injectivity and form.** After establishing that f is bijective, set $y = f(x)$ or $y = x^2$ to simplify. A standard continuation yields $f(x) = x$ (or a translate that verification kills).

Example Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y) - x) = f(xy) \quad \forall x, y \in \mathbb{R}.$$

Set $y = 0$: $f(xf(0) - x) = f(0)$. If $f(0) \neq 1$, the argument $x(f(0) - 1)$ runs over all of $\mathbb{R}$ as x does, therefore f is constant; every constant works. Now assume f is not constant; then necessarily $f(0) = 1$. The equation simplifies to

$$f(x(f(y) - 1)) = f(xy).$$

Further analysis (injectivity of a suitable auxiliary map, or fixing y with $f(y) \neq 1$) produces the linear solutions $f(x) = x + 1$ among others that must be checked by verification.

8. Heuristics — A Practical Solving Checklist

Functional equations are almost never solved by a single flash of insight. They are solved by a short, repeatable sequence of routine moves, followed by one or two more specialised observations. The checklist below is organised in the order you should actually work.

8.1. Stage 1 — Orientation (the first two or three minutes)

1. **Guess a plausible family of solutions.** Test the cheapest candidates: constant functions, $f(x) = kx$, $f(x) = kx + c$, $f(x) = x^2$, the zero function, $f(x) = -x$. A correct guess tells you which properties are worth proving and which constants will have to be determined later. A wrong guess is still useful: it quickly shows you what *cannot* be true.
2. **Absorb obvious freedoms.** If the equation is unchanged when f is shifted or scaled, introduce a new function that removes the freedom. The most common example is the shift to zero:

$$g(x) = f(x) - f(0).$$

The new function g satisfies a cleaner equation and often $g(0) = 0$.

3. **Perform every cheap substitution.** Before any clever idea, systematically try

$$x = 0, \quad y = 0, \quad x = y, \quad y = -x, \quad y = 1, \quad y = f(x).$$

Write down the identity each substitution produces. Even one useful relation can change the whole landscape of the problem.

4. **Determine function at constant values.** Try to take $f(1) = c$, or $f(0) = a$, and then try to find them. After you found, try to get maximum possible information using it.

8.2. Stage 2 — Structural moves

1. **Force cancellation.** Ask yourself: “Which substitution makes many terms identical or equal to zero?” Equating two different arguments of f (the “destroying f ’s” technique) or setting a coefficient to zero frequently collapses a complicated equation into a simple one.
2. **Establish injectivity or surjectivity early.** Look for an isolated variable. Injectivity lets you cancel outer f ’s; surjectivity lets you specialise to convenient values such as $f(x_0) = 0$. Both properties dramatically reduce the number of cases you must consider.
3. **Compare the equation with its swap.** Write $P(x, y)$ and $P(y, x)$. Their difference usually involves only the asymmetric “bumps” and is often the key relation of the whole solution.
4. **Compose the function with itself.** Once you know a relation for $f(f(x))$, apply f once more and equate the two possible associations. The resulting identity frequently involves only a single f .
5. **On $\mathbb{N}$ or $\mathbb{Z}$, manufacture a recurrence.** The substitution $x \mapsto x + 1$ (or simply setting one variable equal to 1) typically produces a linear recurrence that determines every value from a few initial data. After that, ordinary induction finishes the argument.

8.3. Stage 3 — Closing the argument

1. **Avoid the pointwise trap.** It is common to reach a statement such as

$$f(x)^2 = x^2 \quad \text{or} \quad f(x) \in \{0, x\}$$

that holds for each individual x . This does *not* imply that f is globally the identity, or globally zero, or globally $-x$. The choice may still depend on the point x . You must either

prove that the same branch is chosen everywhere, or examine the possible “mixed” functions and eliminate them by a further substitution into the original equation.

2. **Fix all remaining constants.** After the general shape of f is known (for example $f(x) = kx + c$), substitute the expression back into the original equation and see which values of the parameters are admissible.
3. **Verify every candidate.** A function that fails to satisfy the original equation is not a solution, no matter how elegant the derivation that produced it. The verification step takes half a minute and prevents the most frustrating kind of error.

When you feel stuck, return to Stage 1. Ask whether every cheap substitution has really been tried, or whether a different guessed form of f would suggest a different property to prove. The great majority of olympiad functional equations become tractable once the right elementary substitution is found; the sophisticated tricks are needed far less often than beginners expect.

9. Practice Problems

Problems are grouped by the main technique they illustrate. Try each problem yourself before looking for a solution. Remember: always start by guessing a possible form!

9.1. Elementary substitutions

1. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + xf(1-x) = x$$

for all $x \in \mathbb{R}$.

2. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x-y) = f(x) + f(y) - 2xy$$

for all $x, y \in \mathbb{R}$.

3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = x^2y^2 + 2xy$$

for all $x, y \in \mathbb{R}$.

4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) = f(x) + f(y) + f(xy)$$

for all $x, y \in \mathbb{R}$.

5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(a-b)f(a+b) + (b-c)f(b+c) + (c-a)f(c+a) = 0$$

for all $a, b, c \in \mathbb{R}$.

6. Let $a, b \in \mathbb{R}$ be fixed. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y+a) = f(x) + f(y) + b$$

for all $x, y \in \mathbb{R}$.

7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(f(y))) = y + f(f(x))$$

for all $x, y \in \mathbb{R}$.

8. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + f(y)) = f(xy)$$

for all $x, y \in \mathbb{R}$.

9.2. Injectivity and surjectivity

1. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(x + f(y)) = f(x) + y$$

for all $x, y \in \mathbb{Z}$.

2. Find all functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$, where g is injective, such that

$$f(g(x) + y) = g(x + f(y))$$

for all $x, y \in \mathbb{R}$.

3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + y) = 2x + f(f(y) - x)$$

for all $x, y \in \mathbb{R}$.

4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(y + 1)f(x) + f(xf(y) + f(x + y)) = y$$

for all $x, y \in \mathbb{R}$.

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a nonzero function satisfying

$$f(a + 2f(a)f(b)) = f(a) + 2af(b)$$

for all $a, b \in \mathbb{R}$. Prove that f is surjective, and find all such functions.

6. Find all surjective functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y) + y^2) = f((x + y)^2) - xf(x)$$

for all $x, y \in \mathbb{R}$.

7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y) - f(x)) = 2f(x) + xy$$

for all $x, y \in \mathbb{R}$.

8. Let $f : \mathbb{Q}_{>0} \rightarrow \mathbb{Q}$ satisfy

$$f(xy) = f(x) + f(y)$$

for all $x, y \in \mathbb{Q}_{>0}$. Prove that f cannot be injective. Also prove that there exists a surjective function satisfying this equation.

9.3. Destroying f 's and symmetry

1. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + f(y)) = f(x^2) + 2x^2f(y) + f(y)^2$$

for all $x, y \in \mathbb{R}$.

2. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(x) + y) = x + f(y)$$

for all $x, y \in \mathbb{N}$.

3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y)) - x = f(xy)$$

for all $x, y \in \mathbb{R}$.

4. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$(f(a) + b)f(a + f(b)) = (a + f(b))^2$$

for all $a, b \in \mathbb{N}$.

5. Find all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$f(x + 3f(y)) = f(x) + f(y) + 2y$$

for all $x, y \in \mathbb{R}_{>0}$.

6. Let $f, g : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ satisfy $f(1) = g(1)$ and

$$f(g(x) + y) = f(x) + g(y),$$

$$g(f(x) + y) = g(x) + f(y),$$

for all $x, y \in \mathbb{R}_{>0}$. Prove that

$$f(x) = g(x)$$

for all $x \in \mathbb{R}_{>0}$.

7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$x + f(y + f(x)) = y + f(x + f(y))$$

for all $x, y \in \mathbb{R}$, given that the set

$$\left\{ \frac{f(x) - f(y)}{x - y} \mid x, y \in \mathbb{R}, x \neq y \right\}$$

is an interval.

8. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)f(y)) + f(x + y) = f(xy)$$

for all $x, y \in \mathbb{R}$.

9.4. Cauchy-type equations

1. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)^2 + f(y)^2 + f((x + y)^2) = 2x^2 + 2y^2 + f(xf(y) + yf(x))$$

for all $x, y \in \mathbb{R}$.

2. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^{2015} + f(y)^{2015}) = f(x)^{2015} + y^{2015}$$

for all $x, y \in \mathbb{R}$.

3. Find all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ such that

$$f(x) - f(x+y) = f\left(\frac{x}{y}\right) f(x+y)$$

for all $x, y \in \mathbb{R}_{>0}$.

4. Find all monotonic functions

$$f : \mathbb{R} \rightarrow \mathbb{R}_{>0}$$

such that

$$f(x+y) = f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

5. Let $a \in \mathbb{R}$ be fixed. Find all functions

$$f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$$

such that

$$f(x+y) = f(x) + f(y) + a$$

for all $x, y \in \mathbb{R}_{>0}$.

6. Let $d \in \mathbb{R}$ be fixed. Find all continuous functions

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

such that

$$f(x+y+d) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$.

7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + yf(z)) = xf(x) + zf(y)$$

for all $x, y, z \in \mathbb{R}$.

8. Find all functions

$$f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$$

such that

$$f(xf(y)) = yf(x)$$

for all $x, y \in \mathbb{R}_{>0}$ and

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

9.5. Mixed / challenge

1. Find all triples (f, g, h) of injective functions

$$f, g, h : \mathbb{R} \rightarrow \mathbb{R}$$

such that

$$f(x + f(y)) = g(x) + h(y),$$

$$g(x + g(y)) = h(x) + f(y),$$

$$h(x + h(y)) = f(x) + g(y),$$

for all $x, y \in \mathbb{R}$.

2. Find all injective functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for every $x \in \mathbb{R}$ and every positive integer n ,

$$\left| \sum_{i=1}^n i(f(x+i+1) - f(f(x+i))) \right| < 2016.$$

3. Find all functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x - 3f(y)) = xf(y) - yf(x) + g(x)$$

for all $x, y \in \mathbb{R}$, given that

$$g(1) = -8.$$

4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x+y)) = f(yf(x)) + x^2$$

for all $x, y \in \mathbb{R}$.

5. Find all functions

$$f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$$

such that

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy) (f(f(x^2)) + f(f(y^2)))$$

for all $x, y \in \mathbb{R}_{>0}$.

6. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for all $a, b, c \in \mathbb{R}$,

$$a + b + c \geq 0 \implies f(a^3) + f(b^3) + f(c^3) \geq 3f(abc),$$

and

$$a + b + c \leq 0 \implies f(a^3) + f(b^3) + f(c^3) \leq 3f(abc).$$

7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + f(x)f(y)) = xf(x+y)$$

for all $x, y \in \mathbb{R}$.

8. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)f(y)) + f(x+y) = f(xy)$$

for all $x, y \in \mathbb{R}$.

9. Find all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that

$$f(x + f(x)f(y)) = f(x) + xf(y)$$

for all $x, y \in \mathbb{R}_{>0}$.

10. Final Advice for Beginners

1. **Guess first.** Write down the simplest functions that could work (constants, linear, quadratic, $f(x) = 0$, $f(x) = x$, $f(x) = -x$, $\dots$). This gives you a target.
2. **Make elementary substitutions.** $x = 0$, $y = 0$, $x = y$, $y = -x$, $y = 1$, $y = f(x)$, and on $\mathbb{N}$ or $\mathbb{Z}$ also $x \mapsto x + 1$.
3. **Look for isolated variables.** They give injectivity or surjectivity almost for free.
4. **Try to destroy f 's** by equating two arguments.
5. **Exploit symmetry** (and bumps in symmetry) by writing $P(x, y)$ and $P(y, x)$.
6. **Never skip the verification.** After you find a candidate, plug it back in.
7. **If the equation looks additive**, reduce it to Cauchy and remember the plan on $\mathbb{Q}$.
8. **Write cleanly.** Name the original equation $P(x, y)$, then write $P(a, b)$ when you substitute concrete values. This keeps the logic transparent.

Happy solving!

The best way to learn functional equations is to solve many of them.

Sources

The following references formed the basis of the exposition, examples, and practice problems. They were reorganized and expanded with additional explanations aimed at beginners:

- Evan Chen, *Introduction to Functional Equations*
- Amir Hossein Parvardi, *Functional Equations in Mathematical Olympiads*
- Pang-Cheng Wu, Functional Equations handouts and problem sets

Additional examples, rephrasings, and pedagogical remarks were added by the author.